

Data Center FAQs

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Frequently Asked Questions about the Data Center Industry Click on a question below or scroll through this page to view responses to frequently asked questions about data centers. FAQs and Answers What are the economic benefits of data centers? The data center industry provides significant economic benefits for local communities, states, and the U.S. as a whole. In 2024, the U.S. data center industry directly employed more than 1 million workers and supported 5.5 million jobs in total. The industry also provides meaningful tax contributions to local, state, and federal governments that help finance important essential services such as education, infrastructure maintenance (like roads and public transportation), public safety, and health services. In total, the data center industry's tax contribution to local, state, and federal governments was \$204 billion in 2024. How much do local communities benefit from data centers? Counties that have more mature data center markets provide great examples of the many benefits provided by data centers. For example, tax contributions from data centers in Loudoun County, Virginia totaled \$875 million in 2024 and accounted for 38 percent of the county's overall tax revenues—while only comprising 3 percent of the land in Loudoun County. In nearby Prince William County, data centers contributed \$166 million in tax revenue in 2024, a 50% increase over the previous year. More than half of that revenue funds county schools, and the balance goes to general fund priorities, including public safety, parks, libraries, and public transportation. Do data centers benefit other businesses? Data centers enable new innovations and efficiencies across our 21st-century economy. In addition, data centers are vital economic engines for local communities. Data centers create employment opportunities and are catalysts for broader economic growth, supporting ecosystems of suppliers, service providers, and construction. The exponential growth of the data center industry has contributed to the expansion of companies throughout its supply chain. From construction and fabricators of steel used in data center projects, to HVAC manufacturing and portable sanitation companies, the data center industry fuels economic growth in countless companies across a variety of industries. The data center industry has also leaned in, working with manufacturers on shoring up supply chains and helping accelerate project construction timelines. Why do data centers get tax exemptions? Tax exemptions are a common way states attract large economic development projects like data centers and manufacturing. Data centers involve huge financial investments in buildings and equipment, with some individual projects representing billions in capital investment. Given the magnitude of initial capital investment involved and the need for data centers to replace and refresh expensive computer equipment on a regular cycle (typically every 3-5 years), the availability of sales and use tax exemptions on data center equipment is a critical factor for the data center industry in considering markets to locate and expand. More than 35 U.S. states have created economic development programs that recognize the capital-intensive nature of data centers and

treat their equipment with the same standard as manufacturing equipment. These economic development programs for data center equipment are in line with existing programs for other capital-intensive industries like manufacturing, which receive equipment tax exemptions in 41 states. Analyses in both Virginia and Georgia found that, if it weren't for these tax exemption programs, 90% of data center investment would not have occurred in the respective states. This means that the states would miss out on the initial capital investment (often totaling billions per project) and the subsequent annual tax revenues from property taxes, employment taxes, non-exempted sales taxes, etc.—not to mention the continued contributions to state GDP and the job creation in surrounding industries. How many jobs do data centers really create? The exact number of jobs in a data center can vary depending on the company, business model (hyperscale, multitenant, edge, etc.), size of the facility, and other factors, but generally it can range from dozens to hundreds of direct jobs. Plus, a report from PwC finds that each job in the data center industry supports 4.5 jobs elsewhere in the economy. In total, the industry supported 5.5 million jobs in the U.S. in 2024. Do data centers support any other types of jobs? In addition to the well-paying and stable jobs in data center operations, the industry creates many long-term labor and construction jobs. Each individual data center can take multiple years to construct and employs hundreds of workers, with sometimes more than one thousand professionals working at peak construction. With more data center projects being built as campus developments, these construction teams might work on a larger campus site for a decade or longer. Beyond the construction phase, there are dozens of trade jobs in each data center to support ongoing operations and maintenance, including electricians, pipefitters, heating and cooling, and more. For example, IBEW Local 26, the Northern Virginia electrician union, notes that they “have over 30 workers in every single completed data center building doing maintenance work.” For multitenant data centers, in addition to employees who work for the company that owns the data center, there are often dozens of badged employees on site each day that work for the tenant(s). These employees are often not captured in direct employment figures, even though they work in the data center. Are data center jobs good jobs? The data center industry offers high-wage, high-skill jobs that often don't require a four-year degree. Labor income earned directly from the data center industry grew by 11 percent between 2023 and 2024. The increase in labor income earned from the industry has grown even faster than the increase in the number of jobs, suggesting that the U.S. data center industry supports higher-earning jobs at the national level. Why do data centers locate so close to residential areas? Data center companies can only build projects where they are authorized to do so under local zoning ordinances, rules, and regulations. In some instances, land that is zoned for data center use is in proximity to residential areas. In these cases, DCC member companies work with the community to minimize impacts to residents. Data centers strive to be good neighbors in the communities where they operate. More generally, data center operators consider many factors when selecting a location, and access to existing infrastructure like fiber and electricity is a key consideration. That infrastructure often already exists in metropolitan areas, which may also be more desirable for locating data centers because of their proximity to businesses and other end users who rely on data centers for their operations and who require lower latency (i.e., quicker access to data). Are data centers loud? Data center sound, like other industrial facilities, is typically regulated by local government ordinances that require facilities to minimize the volume of operational noise beyond the property lines. A report from Virginia's Joint Legislative Audit & Review Commission (JLARC) found that “data centers emit low-frequency noise that is not loud enough to damage nearby residents' hearing and rarely loud enough to violate noise ordinances.” Furthermore, the data center industry often proactively adopts voluntary noise limits based on established guidance (like Environmental Protection Agency (EPA) or ANSI/ASA S1.1-2013) and invests considerable effort in noise studies, mitigation design, and cooperatively addressing community concerns. Noise challenges, when they arise for specific data centers, often reflect site-specific problems rather than an industry-wide issue. When neighbors raise concerns over noise, data center companies assess the specific situation and employ reasonable mitigation measures, as necessary, to address those concerns. Why do data centers use diesel

backup generators? Data centers need to remain operational during emergencies to ensure that access to essential data and services continues uninterrupted for customers, end users, and the general population – and especially for first responders, hospitals, and government agencies. Due to the critical nature of data center operations, data centers maintain on-site backup generators capable of supporting operations in the event of a utility outage. Some reliability risks, including sudden utility power outages due to storms, natural disasters, and other causes, are inherently outside data center companies' control. Currently, diesel emergency generators are the most commonly deployed technology due to their proven reliability, availability in the supply chain, siting flexibility, and workplace safety profile. Backup diesel generators are highly-regulated resources that are typically used for a maximum of 30 minutes in an average month. They are typically only used for brief maintenance and during electricity outages to ensure continuity and digital services provided by data centers, a fact that is underscored by their limited use. While there are no feasible alternatives available today in the market to replace traditional backup generators at scale, data centers are actively evaluating new technologies that can provide similar reliability, fuel availability, siting flexibility, and workplace safety protections. As an interim measure, many data centers are looking to low carbon fuel alternatives, including renewable diesel, where available. Do data centers use a lot of water? The amount of water a data center uses depends on the technology each facility employs to cool its servers—air cooling, water cooling, or some combination. There is no one-size-fits-all solution for cooling, and data center operators take into account local factors when determining which cooling technology to use. These factors include humidity, climate/temperature, and availability of water, including recycled, non-potable, or harvested rainwater sources. How much water is used by data centers that employ water cooling technology? This varies depending on the size of the facility, climate and outdoor temperature, the type of compute done on the servers, and other factors. A recent report from an independent government commission in Virginia found that 83% of data centers in Virginia—the largest data center market in the world—use the same amount of water (or less) as a large office building. For context, the average large office building in Virginia uses 6.7 million gallons of water per year (or approximately 18,400 gallons per day—the amount needed to fill an average residential swimming pool). What is the industry doing to reduce its impact on local water resources? Data center companies prioritize efficient water practices in operations and development. The industry is actively investing in and deploying innovations, such as waterless cooling systems, closed-loop cooling systems, and the use of recycled or reclaimed water. In addition, local water availability is a key consideration in the cooling systems and strategies data centers choose to employ. Data centers also actively engage in water and watershed conservation and restoration projects in local communities and globally. Why do data centers use so much energy? There is unprecedented demand for the digital services that have become central to our daily lives and the modern economy. These digital, cloud-based services on which we all rely take place in physical locations—America's data centers. With an average of 21 connected devices per household in the U.S. and 5.5 billion people currently online globally, the role of data centers is expected to grow, as consumers and businesses generate twice as much data in the next five years as they did in the past decade. This growth is driven by the widespread adoption of cloud services, the proliferation of connected devices, and the rapid scaling of advanced technologies like generative AI. Data centers are like the brains of the internet. They process, store, and communicate all the data behind the digital services we rely on every day—everything from online shopping and streaming movies to financial transactions, government services and telehealth appointments. The unprecedented increase in demand for these digital services is driving growth in the data center industry. To perform these functions, data centers are equipped with servers and other devices that require a significant amount of energy to operate. These devices generate heat while in operation, and power also is required to remove heat and maintain operating temperatures in many data centers. It's important to recognize that data centers aren't the only industry that is experiencing increases in energy use. After nearly two decades of flat electricity demand growth, the U.S. is now experiencing a need for more power, driven by the onshoring of new manufacturing, widespread

electrification of buildings, industry, and transportation, hydrogen fuel production, and the digitization of our economy. Is the data center industry doing anything about its energy use? While managing the electrical grid is the role of utilities, grid operators, and regulators, data center companies are leaning in as engaged partners across the country to ensure both continued data center development and an affordable, reliable electricity grid for all customers. The data center industry has been a leader in expanding access to clean energy and remains committed to sustainability and innovation. In fact, according to S&P Global Insight, U.S. data centers accounted for half of total U.S. corporate clean energy procurement through the third quarter of 2024. DCC member companies are looking beyond the energy challenges of this moment. This can be seen through announcements and investments in everything from small modular reactors (SMRs) and enhanced geothermal to long-duration battery storage and carbon capture/sequestration. How do data centers unlock greater efficiency for our modern society? Data centers aggregate our collective computing demands efficiently and securely. Previously, these types of computing resources were dispersed across businesses, which was far less efficient and secure. In 2010, nearly 80 percent of data center computing was conducted in smaller traditional computer centers. By 2018, approximately 89 percent of data center computing took place in larger cloud data centers. By centralizing computing resources, data centers leverage innovations in design, equipment, and technology to maximize energy efficiency. While electricity consumption at data centers rose 6 percent from 2010 to 2018, computing output jumped 550 percent, marking significant gains in efficiency and productivity. It is important to recognize data centers are not simply large consumers of electricity; they also facilitate efficiency gains for homes, businesses, industrial consumers, and utilities across the economy. Many technologies and strategies deployed across the country—including smart thermostats, smart meters, managed electric vehicle charging, smart lighting, and grid enhancement technologies—require the digital infrastructure provided by data centers. The U.S. Department of Energy recently released a report identifying the ways in which AI applications supported by data centers will increasingly and significantly enhance the way in which our electricity grid is operated, particularly by enabling better grid planning and forecasting, streamlining siting and permitting processes, and improving grid reliability and system reliance. Are data centers paying their fair share of electricity costs? DCC's member companies are committed to paying their full cost of service for the energy they use. Data center companies routinely pay utilities for infrastructure directly associated with serving data center facilities, which can include breakers, transformers, entire substations, and other supporting infrastructure. A 2026 report from energy consulting firm E3 found no historical evidence that data centers are driving increases in residential electricity costs under existing rate structures. In fact, adding data centers to the grid can help put downward pressure on rates for other electricity users. In Virginia, the Joint Legislative Audit and Review Commission (JLARC) found that "current rates appropriately allocate costs to the customers responsible for incurring them, including data center customers." While the report emphasizes a need to continue reviewing current practices and consider alternative approaches as system growth continues, it is important to note that the existing cost allocation and rate review practices have worked to date. The same has also been found in other critical data center markets like Arizona, where the Arizona Corporation Commission (ACC) stated that "To date in Arizona, the data center industry is paying its fair share of expansion of power generation and infrastructure under existing utility rate structures and Commission practices."